

Semester	B.E. Semester VII – Computer Engineering
Subject	R programming
Subject In-charge	Prof. SDW

Student Name	Diksha Parulekar.
Roll Number	23102A0068.
Grade and Subject Teacher's Signature	

Experiment Number	06	
Experiment Title	Customer Segmentation and Predictive Analytics Using Machine Learning	
Resources / Apparatus Required	Hardware: Computer system	Software: Python
Description	Github : dikshaparulekar/23102A0068_R-PROGRAMMING This experiment builds an end-to-end customer analytics solution for an e-commerce company using the UCI Online Retail dataset. The dataset has 541,909 transactions from a UK-based online retailer between December 2010 and December 2011. It records the invoice number, product, quantity, invoice date, unit price, customer ID and country. The goals are to group customers into meaningful segments and to predict which customers belong to the high-value segment. The data was first cleaned by removing duplicate rows, records with a missing CustomerID, cancelled invoices, and lines with non-positive quantity or price. The transactions were then summarised into one row per customer. The features were Recency, Frequency, Monetary value, average transaction value, total quantity purchased and purchase frequency. Because these features are heavily right-skewed, they were log-transformed, capped using the IQR rule to limit the influence of outliers, and standardised so that every feature contributes equally to distance-based algorithms. K-Means clustering was applied, and the Elbow Method (supported by average silhouette widths) was used to choose the number of clusters. The result was compared with Ward hierarchical clustering, using a dendrogram, the Silhouette Score and the Adjusted Rand Index. Principal Component Analysis reduced the data for a 2D projection and an interactive 3D Plotly visualisation, which showed how well the segments separate. Each cluster was profiled on its original, unscaled values	

and given a business label: Champions, Potential Loyalists, At-Risk Customers, or Lost / Low-Value.

The highest-spending segment was defined as the high-value class. Random Forest and SVM (RBF kernel) classifiers were trained on a stratified 80/20 split to predict membership of this class. They were evaluated on Accuracy, Precision, Recall, F1-Score, ROC-AUC, confusion matrices and ROC curves. Feature importance was analysed to see which behaviours drive the prediction. The experiment closes with marketing strategies tailored to each segment.

**Program
&
Output:**

The screenshot displays the RStudio interface for a file named '23102A0068_Diksha Parulekar_R programming assignment 6.ipynb'. The editor shows R code for loading and inspecting the 'UCI Online Retail' dataset. The console output provides a detailed overview of the data, including variable types, missing values, and data quality indicators.

```

out[order(out$InvoiceDate), ]

raw <- load_uci()
if (is.null(raw)) {
  message("UCI download unavailable -> using synthetic fallback dataset with the same schema.")
  raw <- make_synthetic_retail()
  data_source <- "Synthetic fallback (UCI Online Retail schema)"
} else {
  data_source <- "UCI Online Retail (real data)"
}
cat("Data source :", data_source, "\n")
cat("Rows x Cols :", nrow(raw), "x", ncol(raw), "\n")
head(raw)

```

Data source : UCI Online Retail (real data)
Rows x Cols : 541909 x 8

A data frame: 6 x 8

InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
<chr>	<chr>	<chr>	<dbl>	<dtm>	<dbl>	<dbl>	<chr>
1	536365	85123A WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
2	536365	71053 WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
3	536365	84406B CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
4	536365	84029G KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
5	536365	84029E RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
6	536365	22752 SET 7 BABUSHKA NESTING BOXES	2	2010-12-01 08:26:00	7.65	17850	United Kingdom

5. Exploratory Overview of the Raw Data

```

---- Structure ----
'data.frame':   541909 obs. of  8 variables:
 $ InvoiceNo  : chr  "536365" "536365" "536365" "536365" ...
 $ StockCode : chr  "85123A" "71053" "84406B" "84029G" ...
 $ Description: chr  "WHITE HANGING HEART T-LIGHT HOLDER" "WHITE METAL LANTERN" "CREAM CUPID HEARTS COAT HANGER" "KNITTED UNION FLAG HOT WATER BOTTLE" ...
 $ Quantity  : num  6 6 8 6 6 2 6 6 32 ...
 $ InvoiceDate: POSIXct, format: "2010-12-01 08:26:00" "2010-12-01 08:26:00" ...
 $ UnitPrice : num  2.55 3.39 2.75 3.39 3.39 7.65 4.25 1.85 1.85 1.69 ...
 $ CustomerID: num  17850 17850 17850 17850 17850 ...
 $ Country   : chr  "United Kingdom" "United Kingdom" "United Kingdom" "United Kingdom" ...

---- Missing values per column ----
Column Missing Percent
1 InvoiceNo      0 0.00
2 StockCode     0 0.00
3 Description    1454 0.27
4 Quantity      0 0.00
5 InvoiceDate    0 0.00
6 UnitPrice     0 0.00
7 CustomerID    135080 24.93
8 Country       0 0.00

---- Data quality indicators ----
Cancelled-invoice lines : 9288 (1.71%)
Lines with Quantity <= 0 : 10624
Lines with UnitPrice <= 0 : 2517
Date range : 2010-12-01 08:26:00 to 2011-12-09 12:58:00
Unique invoices : 25900
Unique customers : 4372
Countries : 38
Quantity      UnitPrice
Min. : -80995.000 Min. : -11062.060
1st Qu.: 1.000   1st Qu.: 1.250
Median : 3.000   Median : 2.080
Mean : 9.552    Mean : 4.611
3rd Qu.: 10.000 3rd Qu.: 4.130
Max. : 80995.000 Max. : 38970.000

```

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```
[ ] head(customers)
cat("\n")
print(round(sapply(customers[, c(cluster_vars, extra_vars)], function(x) c(Min = min(x), Median = median(x), Mea
```

*** Snapshot date: 2011-12-10
Customers: 4338

A data frame: 6 × 10

	CustomerID	Recency	Frequency	Monetary	TotalQuantity	UniqueProducts	TenureDays	AvgTransactionValue	PurchaseFrequency
	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<int>	<dbl>	<dbl>	<dbl>
1	12346	326	1	77183.60	74215	1	326	77183.6000	
2	12347	3	7	4310.00	2458	103	368	615.7143	
3	12348	76	4	1797.24	2341	22	359	449.3100	
4	12349	19	1	1757.55	631	73	19	1757.5500	
5	12350	311	1	334.40	197	17	311	334.4000	
6	12352	37	8	2506.04	536	59	297	313.2550	

	Recency	Frequency	Monetary	AvgTransactionValue	TotalQuantity
Min	1.00	1.00	3.75	3.45	1.00
Median	51.00	2.00	668.57	291.94	378.00
Mean	93.06	4.27	2048.69	417.65	1187.64
Max	374.00	209.00	280206.02	84236.25	196915.00

	PurchaseFrequency	UniqueProducts	CancelCount	TenureDays
Min	0.08	1.0	0.00	1.00
Median	0.45	35.0	0.00	249.00
Mean	0.63	61.5	0.83	223.83
Max	16.76	1787.0	47.00	374.00

8. Outlier Treatment and Feature Scaling

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```
[ ] long_raw$Feature <- factor(long_raw$Feature, levels = cluster_vars)

ggplot(long_raw, aes(x = Feature, y = Value, fill = Feature)) +
  geom_boxplot(outlier.colour = "red", outlier.alpha = 0.4, show.legend = FALSE) +
  facet_wrap(~Feature, scales = "free", nrow = 2) +
  labs(title = "Raw RFM features: heavy right skew and outliers (red points)", x = NULL, y = NULL) +
  theme(axis.text.x = element_blank())
```

Raw RFM features: heavy right skew and outliers (red points)

```
[ ] # ---- Treatment functions ----
cap_iqr <- function(x) {
  q <- quantile(x, c(0.25, 0.75), na.rm = TRUE)
  iqr <- q[2] - q[1]
  if (iqr == 0) return(x)
  pmin(pmax(x, q[1] - 1.5 * iqr), q[2] + 1.5 * iqr)
}
```

Variables Terminal

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Correlation between engineered features:

...

	Recency	Frequency	Monetary	AvgTransactionValue
Recency	1.00	-0.57	-0.49	-0.13
Frequency	-0.57	1.00	0.81	0.17
Monetary	-0.49	0.81	1.00	0.70
AvgTransactionValue	-0.13	0.17	0.70	1.00
TotalQuantity	-0.49	0.76	0.93	0.64
PurchaseFrequency	-0.73	0.60	0.48	0.12

TotalQuantity PurchaseFrequency

Recency	-0.49	-0.73
Frequency	0.76	0.60
Monetary	0.93	0.48
AvgTransactionValue	0.64	0.12
TotalQuantity	1.00	0.48
PurchaseFrequency	0.48	1.00

After log transform, IQR capping and standardisation

Recency

Frequency

Monetary

AvgTransactionValue

TotalQuantity

PurchaseFrequency

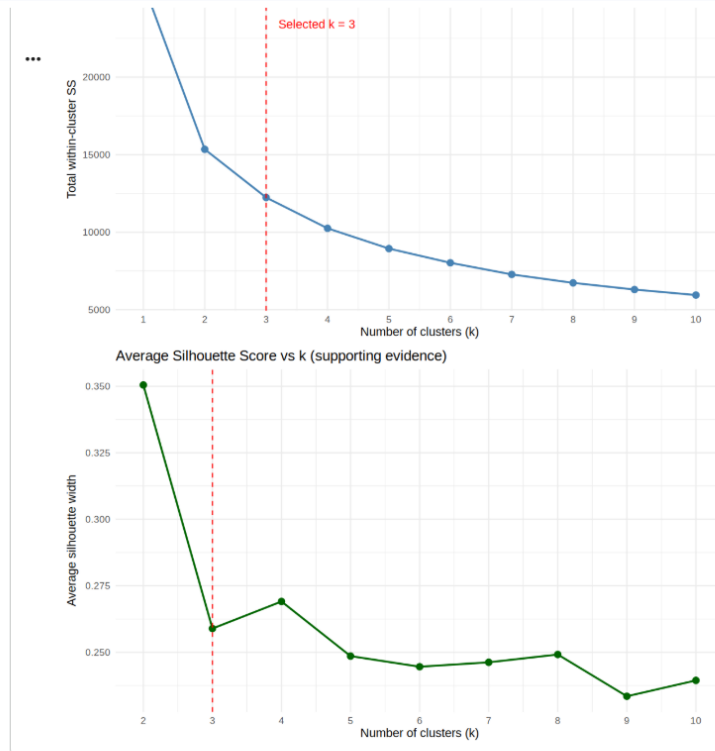
Variables Terminal



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{ } Variables [] Terminal



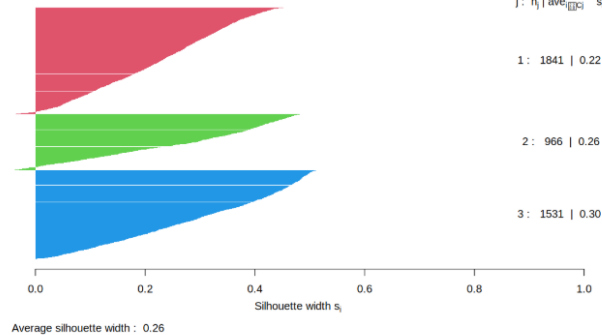
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K-Means with $k = 3$
Cluster sizes:
1 2 3
1841 966 1531
Between-SS / Total-SS: 52.99 %
Average Silhouette Score (K-Means): 0.2589

Silhouette plot: K-Means ($k = 3$)
 $n = 4338$



11. Hierarchical Clustering and Dendrogram

Agglomerative clustering with **Ward's method** (`ward.D2`) on the same standardised features. The tree is cut at the same k as the K-Means method. The results of the two methods can be compared fairly.

+ Code + Text

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```
ggplot(comparison, aes(Method, Avg_Silhouette, fill = Method)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = Avg_Silhouette), vjust = -0.4) +
  labs(title = "Average Silhouette Score: K-Means vs Hierarchical", y = "Average silhouette width", x = NULL) +
  ylim(0, max(comparison$Avg_Silhouette) * 1.2)
```

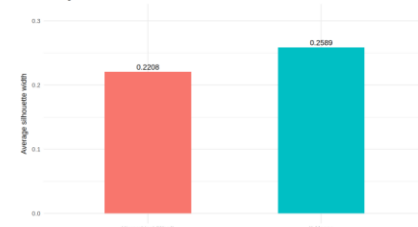
Method	Clusters	Avg_Silhouette	Smallest_Cluster	Largest_Cluster
K-Means	3	0.2589	966	1841
Hierarchical (ward)	3	0.2208	1813	1857

Adjusted Rand Index (agreement between K-Means and Hierarchical): 0.4226

Cross-tabulation (rows = K-Means, columns = Hierarchical):

KMeans	Hierarchical		
	1	2	3
1	1310	504	19
2	11	954	1
3	528	10	993

Higher silhouette: K-Means -> K-Means is used for profiling and for the supervised models.
(Silhouette interpretation: >0.5 strong, 0.25-0.5 reasonable, <0.25 weak structure.)
Average Silhouette Score: K-Means vs Hierarchical



Variables Terminal

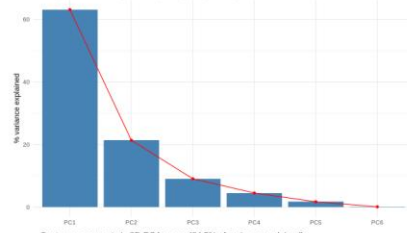
23102A0068_Diksha Parulekar_R programming assignment 6.ipynb

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```
avg_transactionvalue 0.298 0.048 0.363
TotalQuantity 0.472 0.231 0.163
PurchaseFrequency 0.365 0.471 0.359
```

Scree plot: variance explained by each principal component



Customer segments in 2D PCA space (84.5% of variance explained)



Variables Terminal

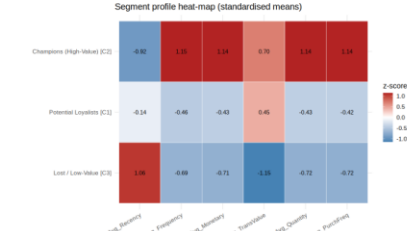
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```
Segment Customers Pct_Customers Pct_Revenue Avg_Recency
Champions (High-Value) [C2] 966 22.27 71.88 18.86
Potential Loyalists [C1] 1841 42.44 24.64 74.68
Lost / Low-Value [C3] 1531 35.29 4.28 162.58
Avg_Frequency Avg_Monetary Avg_TransValue Avg_Quantity Avg_PurchaseFreq
11.55 6539.58 579.31 3766.65 1.35
2.77 1189.28 525.74 786.93 8.58
1.49 248.55 185.66 138.45 0.34
```

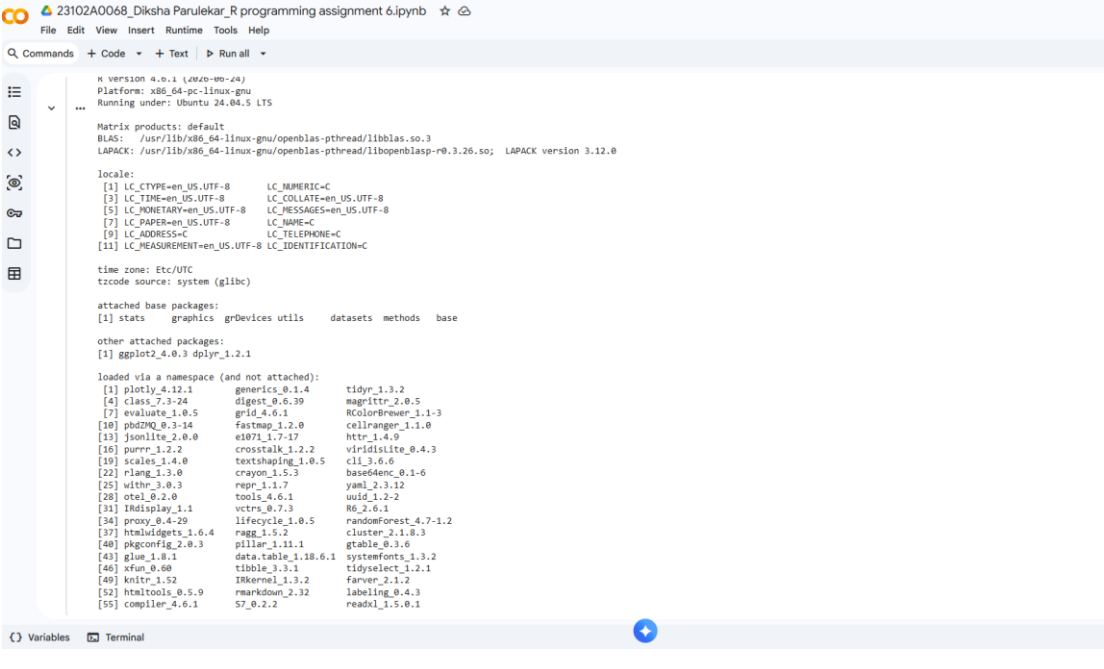
Segment profile heat-map (standardised means)



Share of customers vs share of revenue by segment



Variables Terminal

	 The screenshot shows a Jupyter Notebook environment with the following details: Kernel: Python 3.12.0 Platform: x86_64-pc-linux-gnu Running under: Ubuntu 24.04.5 LTS Matrix products: default BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3 LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p-r0.3.26.so; LAPACK version 3.12.0 locale: [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C [3] LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8 [5] LC_MESSAGES=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8 [7] LC_PAPER=en_US.UTF-8 LC_NAME=C [9] LC_ADDRESS=C LC_TELEPHONE=C [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C time zone: Etc/UTC tzcode source: system (glibc) attached base packages: [1] stats graphics grDevices utils datasets methods base other attached packages: [1] ggplot2_4.0.3 dplyr_1.2.1 loaded via a namespace (and not attached): [1] plotly_4.12.1 generics_0.1.4 tidyrr_1.3.2 [4] class_7.3-24 digest_0.6.39 magrittr_2.0.5 [7] evaluate_1.0.5 grid_4.6.1 RColorBrewer_1.1-3 [10] pbdfzmq_0.3-14 fastmap_1.2.0 cellranger_1.1.0 [13] jsonlite_2.0.0 e1071_1.7-17 httr_1.4.9 [16] purrr_1.2.2 crosstalk_1.2.2 viridisLite_0.4.3 [19] scales_1.4.0 textshaping_1.0.5 cli_3.6.6 [22] rlang_1.3.0 crayon_1.5.3 base64enc_0.1-6 [25] withr_3.0.3 repr_1.1.7 yaml_2.3.12 [28] otl_0.2.0 tools_4.6.1 uuid_1.2-2 [31] IRdisplay_1.1 vctrs_0.7.3 R6_2.6.1 [34] proxy_0.4-29 lifecycle_1.0.5 randomForest_4.7-1.2 [37] htmlwidgets_1.6.4 ragg_1.5.2 cluster_2.1.8.3 [40] pkgconfig_2.0.3 pillar_1.11.1 gttable_0.3.6 [43] glue_1.8.1 data.table_1.18.6.1 systemfonts_1.3.2 [46] xfun_0.60 tibble_3.3.1 tidyselct_1.2.1 [49] knitr_1.52 IRkernel_1.3.2 farver_2.1.2 [52] htmltools_0.5.9 rmarkdown_2.32 labeling_0.4.3 [55] compiler_4.6.1 S7_0.2.2 readxl_1.5.0.1
Conclusion:	The experiment shows that raw transaction logs can be turned into actionable customer intelligence. Careful preprocessing, especially handling cancellations, missing customer IDs, skewness and outliers, was essential for producing stable, interpretable clusters. The Elbow Method and silhouette analysis indicated natural customer segments. K-Means achieved an average Silhouette Score of against for hierarchical clustering. The Adjusted Rand Index shows that the two methods broadly agree on the structure. PCA showed that the first two components capture % of the variance, and the segments are clearly visible in both the 2D and the interactive 3D projections. Profiling showed a strong concentration of value. The high-value segment makes up only of customers but generates of revenue, while the lowest-value segment is large in headcount but contributes very little revenue. This supports focusing the marketing budget on the segments that matter most. Both supervised models predicted high-value membership accurately. Random Forest reached an F1-Score of and ROC-AUC of and SVM reached so performed slightly better. The label is derived from clusters built on the same behavioural features, so these high scores are expected. The classifiers are best seen as a fast scoring rule that can place new customers into the high-value segment without re-running the clustering. Feature importance showed that are the strongest indicators of a high-value customer.